



4026 - Hot dust in the nearby metal-poor starburst galaxy CGCG 007-025: A window to the high redshift Universe

Cycle: 2, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1		MIRI Medium Resolution Spectroscopy	(1) CGCG007-025
	2		MIRI Medium Resolution Spectroscopy	(2) CGCG007-025-BACKGROUND

ABSTRACT

Although rare in the nearby Universe, metal-poor starbursting dwarf galaxies are promising sources to be used as benchmarks for the study of the ISM in high- z galaxies. Here we propose MIRI-MRS IFU observations to characterise in detail the dust emission in one of the most compelling local analogues, CGCG007-025. This young, compact, metal-poor starburst is powered by massive stars (including carbon Wolf-Rayets) and features a steeply rising mid-IR SED, indicative of the presence of hot dust. With the enhanced sensitivity and the superb spatial resolution enabled by JWST, we will be able to pinpoint the exact location of the dust emission and characterise it both spatially and spectrally. We will infer the abundances of

heavy elements from mid-IR fine-structure lines and compare it with the (possibly biased) optical diagnostics; we will place constraints on the hardness of the radiation field and the extent of the density-bounded region; and we will look for faint PAH emission and search for the elusive trace of silicates. These observations will provide a detailed characterisation of the physical properties of dust under the extreme conditions that were common in primordial ISMs--but with a degree of detail and sensitivity unmatched by any observation of the high redshift Universe.

OBSERVING DESCRIPTION

We propose MIRI-MRS observations of the nearby starburst dwarf galaxy CGCG007-025 to carry out a detailed characterisation of the dust emission in the most compact star-forming clump.

The 5-28 um mid-IR IFU spectra will be used to (1) detect emission lines from all ionic species; (2) look for the presence of (yet undetected) PAH features; (3) search for possible broad continuum features from silicates.

The exposure time has been estimated to ensure that detection of continuum features will be possible even at the blue end of the MRS spectra. We adopt an 'extended source' 4pt dither strategy, and also observe an adjacent background field to the south of our target.

To improve the MRS astrometric accuracy we employ MIRI simultaneous imaging. We propose to check if it would be possible to image the target galaxy while the background field is observed with the IFU--but only if this has no impact on the visibility window.

Proposal 4026 - Targets - Hot dust in the nearby metal-poor starburst galaxy CGCG 007-025: A window to the high redshift Universe

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	CGCG007-025	RA: 09 44 1.8800 (146.0078333d) Dec: -00 38 32.09 (-.64225d) Equinox: J2000		
	Comments: Category=Galaxy Description=[Dwarf galaxies, Emission line galaxies] Extended=YES				
Fixed Targets	(2)	CGCG007-025- BACKGROUND	RA: 09 44 1.8800 (146.0078333d) Dec: -00 39 0.61 (-.65017d) Equinox: J2000		
	Comments: Category=Calibration Description=[Telescope/sky background]				

Proposal 4026 - Observation 1 - Hot dust in the nearby metal-poor starburst galaxy CGCG 007-025: A window to the high redshift Uni...

Observation	Proposal 4026, Observation 1													Thu May 11 13:09:26 GMT 2023
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
	Background Observations:[Observation 2]													
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name	Target Coordinates					Targ. Coord. Corrections			Miscellaneous			
	(1)	CGCG007-025	RA: 09 44 1.8800 (146.0078333d) Dec: -00 38 32.09 (-.64225d) Equinox: J2000											
	Comments: Category=Galaxy Description=[Dwarf galaxies, Emission line galaxies] Extended=YES													
Acquisition	#													Target
	1													NONE
Template	AcqFilter	Primary Channel					Simultaneous Imaging			Imager Subarray				
		ALL					YES			FULL				
Dithers	#	Dither Type					Optimized For			Direction				
	1	4-Point					EXTENDED SOURCE			NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1		IMAGER	F770W	FASTR1	50	2	1	Dither 1	4	8	1121.116		
	1	SHORT(A)	MRSLONG		FASTR1	50	2	1	Dither 1	4	8	1121.116		
	1	SHORT(A)	MRSSHORT		FASTR1	50	2	1	Dither 1	4	8	1121.116		
	2		IMAGER	F1000W	FASTR1	50	2	1	Dither 1	4	8	1121.116		
	2	MEDIUM(B)	MRSLONG		FASTR1	50	2	1	Dither 1	4	8	1121.116		
	2	MEDIUM(B)	MRSSHORT		FASTR1	50	2	1	Dither 1	4	8	1121.116		
	3		IMAGER	F560W	FASTR1	50	2	1	Dither 1	4	8	1121.116		
	3	LONG(C)	MRSLONG		FASTR1	50	2	1	Dither 1	4	8	1121.116		
	3	LONG(C)	MRSSHORT		FASTR1	50	2	1	Dither 1	4	8	1121.116		

Special Requirements	Sequence Observations 1, 2, Non-interruptible
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Proposal 4026 - Observation 2 - Hot dust in the nearby metal-poor starburst galaxy CGCG 007-025: A window to the high redshift Uni...

Observation	Proposal 4026, Observation 2													Thu May 11 13:09:26 GMT 2023
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
	Background Observation For: [Observation 1]													
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name	Target Coordinates					Targ. Coord. Corrections			Miscellaneous			
	(2)	CGCG007-025-BACKGROUND	RA: 09 44 1.8800 (146.0078333d) Dec: -00 39 0.61 (-.65017d) Equinox: J2000											
Acquisition	Comments: Category=Calibration Description=[Telescope/sky background]													
	#													Target
	1													NONE
Template	AcqFilter	Primary Channel					Simultaneous Imaging			Imager Subarray				
		ALL					YES			FULL				
Dithers	#	Dither Type					Optimized For			Direction				
	1	2-Point					EXTENDED SOURCE			NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1		IMAGER	F770W	FASTR1	50	2	1	Dither 1	2	4	560.558		
	1	SHORT(A)	MRSLONG		FASTR1	50	2	1	Dither 1	2	4	560.558		
	1	SHORT(A)	MRSSHORT		FASTR1	50	2	1	Dither 1	2	4	560.558		
	2		IMAGER	F1000W	FASTR1	50	2	1	Dither 1	2	4	560.558		
	2	MEDIUM(B)	MRSLONG		FASTR1	50	2	1	Dither 1	2	4	560.558		
	2	MEDIUM(B)	MRSSHORT		FASTR1	50	2	1	Dither 1	2	4	560.558		
	3		IMAGER	F560W	FASTR1	50	2	1	Dither 1	2	4	560.558		
	3	LONG(C)	MRSLONG		FASTR1	50	2	1	Dither 1	2	4	560.558		
	3	LONG(C)	MRSSHORT		FASTR1	50	2	1	Dither 1	2	4	560.558		

Special Requirements	Sequence Observations 1, 2, Non-interruptible
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